

Question 1 - 0417/May/June/24/11 Q12

One of the threats to personal data is credit card cloning.

(a) Explain what is meant by credit card cloning.

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[3]

(b) One security improvement for this type of threat is the use of contactless cards.

Give **two** other security improvements that could be made to help reduce this type of threat.

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2

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[2]

Question 2 - 0417/May/June/24/12 Q2

Complete each of the following sentences which relate to health and safety issues when using ICT equipment.

(a) Trailing leads in a computer room can cause

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[1]

(b) Continual typing on a keyboard can cause

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[1]

(c) Staring at a computer screen for long periods of time can cause

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[1]

(d) Spilling water onto computer equipment can cause

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[1]

Question 3 - 0417/May/June/24/12 Q15

Explain why data protection legislation is required.

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[4]

Question 4 - 0417/May/June/24/11 Q2

Complete each of the following sentences.

(a) The scrambling of data being sent from one device to another device is called
..... [1]

(b) Using a computer to maintain the temperature of a room is called
..... [1]

(c) Unauthorised access to a computer system is called
..... [1]

(d) A conversation between a systems analyst and a member of staff that is part of the analysis
is called
..... [1]

Question 5 - 0417/May/June/24/11 Q7

An organisation has provided laptops for its sales team to use when working at home and in the office.

- (a) Give **two** benefits to the employees of using these laptops for their work.

Benefit 1

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Benefit 2

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[2]

- (b) Give **two** issues that may arise from providing employees with these laptops.

Issue 1

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Issue 2

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[2]

- (c) The network manager of the organisation has been asked to record the details of the employees who have been provided with laptops. The network manager has decided to produce an online data capture form.

Describe the main design features that should be considered to make the online data capture form easy to complete accurately.

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[4]

- (d) The employees are given a list of strategies to follow to reduce the health risks of using laptops.

State **four** strategies that could be included in the list.

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2
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3
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4
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[4]

Question 6 - 0417/May/June/24/11 Q3

Roger is a 16-year-old student who is setting up a password for his computer. He has four ideas for the passwords he may use.

- (a) Select the most appropriate password from the list and give reasons for your choice.

Roger2008

R08123

RogerZpa55w0rd

6lRrg08&

Password

Reasons

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.....
.....

[3]

- (b) State **one** way of avoiding password interception.

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[1]

Question 7 - 0417/May/June/24/13 Q7

A teacher is creating a revision website for her students. The website contains information taken from a number of different sources including published books. These sources are protected by copyright.

- (a) State what the teacher needs to do to ensure that she does not break copyright rules when she publishes the website.

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..... [2]

- (b) Explain the principles of software copyright.

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..... [3]

- (c) Explain why copyright legislation is needed.

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..... [3]

Question 8 - 0417/May/June/24/13 Q12

One of the risks when entering a password into a computer system is key logging.

(a) Explain what is meant by key logging.

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[2]

(b) Give **two** security improvements that could be made to prevent key logging.

1

2

[2]

Question 9 - 0417/May/June/24/12 Q12

One of the risks when entering a password into a system is shoulder surfing.

(a) Explain what is meant by shoulder surfing.

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[2]

(b) Give **two** security improvements that could be made to reduce shoulder surfing.

1

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2

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[2]

Question 10 - 0417/May/June/24/12 Q3

Remotely logging in to a college computer system requires two-factor authentication.

(a) Explain what is meant by two-factor authentication.

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[2]

(b) State **two** examples of methods that could be used as part of two-factor authentication.

Example 1

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Example 2

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[2]

Question 11 - 0417/May/June/24/13 Q3

A student is able to access a secure area of the school's website using zero login.

(a) Explain what is meant by zero login.

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[2]

(b) Biometrics could be used as another method of accessing the secure area.

State **three** examples of biometric methods.

1

2

3

[3]

Question 12 - 0417/Oct/Nov/23/13 Q8

Many devices in the home can be connected to the internet so they can be controlled remotely. These devices are called smart devices. A house contains a smart security system.

(a) Describe the benefits and drawbacks of using a smart security system.

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[4]

(b) Identify **two** sensors that could be used with the smart security system.

1

2

[2]

(c) Part of the smart security system involves an automatic gate for vehicles approaching the house. There is a CCTV camera at the gate and the smart security system has software for recognising each vehicle's number plate.

Explain how automatic number plate recognition (ANPR) systems work.

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[4]

(d) Describe **three** problems that could occur when the ANPR system reads a vehicle number plate.

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[3]

Question 13 - 0417/Oct/Nov/23/11 Q7

An employee in a company has used QWERTY123 as his login password for some time.

(a) Give **three** reasons why this is **not** advisable.

- 1
- 2
- 3

[3]

(b) Complete the sentences using the most appropriate data protection method from the list.
Your answers must be different in each case.

Anti-spyware Biometrics A digital certificate A dongle
Encryption A firewall Two-factor authentication User ID

- (i) A method which scrambles the data so that it cannot be understood is
..... [1]
- (ii) A fingerprint is an example of
..... [1]
- (iii) A method of guaranteeing a website is authentic is
..... [1]
- (iv) A hardware device that filters the information coming through the Internet connection into a LAN is
..... [1]

(c) An employee in the company uses his company password for other accounts.

Explain why this is **not** a good idea.

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..... [1]

Question 14 - 0417/Oct/Nov/23/11 Q4

A headteacher is considering setting up a card system for students to buy food in the school canteen.

The headteacher is trying to decide whether to install a system that uses magnetic stripe card readers or one that uses chip and PIN card readers.

Compare the use of magnetic stripe card readers with the use of chip and PIN card readers. Your answer must include similarities and differences.

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..... [6]

Question 15 - 0417/May/June/23/11 Q5

State the safety issue that is most likely to occur for each of these causes.

(a) Spilling drinks on computer equipment

..... [1]

(b) Overloading sockets by plugging in too many devices

..... [1]

(c) Trailing cables across the floor

..... [1]

(d) Placing a laser printer at the edge of a table

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..... [1]

Question 16 - 0417/Oct/Nov/23/13 Q4

A company's computer system is protected by both a user ID and a password.

- (a)** Explain how a user ID and a password are used to increase the security of data.

[illegible]

- (b)** The company plans to improve the security of its computer system by using biometric methods.

Describe what is meant by biometric methods. Give **two** examples of biometrics.

Description

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Example 1

Example 2

[4]

Question 17 - 0417/May/June/23/11 Q6

A student has set up a password for his laptop computer. The password he has chosen is Px2zY!Cg3?bd

- (a)** Discuss the benefits and drawbacks of using this password.

[illegible]

- (b)** The student could have protected his laptop computer by using two-factor authentication. This could have combined his password with a biometric method.

State **two** examples of biometric methods he could use.

1

2

[2]

Question 18 - 0417/May/June/23/13 Q5

Identify **two** appropriate IT-related causes for each of the health issues. Your answers must be different in each case.

(a) repetitive strain injury (RSI)

- 1
- 2

[2]

(b) eye strain

- 1
- 2

[2]

Question 19 - 0417/May/June/23/13 Q6

Two-factor authentication can use a number of different forms of identification.

(a) One form of identification is to use 'where the user is'.
An example of this would be the user's GPS location.

State **two** other forms of identification. Give an appropriate example of each.

- 1
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Example

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- 2
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Example

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[4]

(b) Explain the purpose of two-factor authentication.

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[2]

Question 20 - 0417/May/June/23/12 Q5

State which health issue is most likely to occur from each description shown. Your answer must be different in each case.

(a) typing on a keyboard for long periods of time

..... [1]

(b) looking at a computer screen for a long time

..... [1]

(c) sitting with poor posture

..... [1]